CHAPTER 8: PROJECT MANAGEMENT

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A INTRODUCTION TO PROJECT MANAGEMENT

PROJECT MANAGEMENT

- Project management is the **application of processes, methods, skills, knowledge and experience** to achieve **specific project objectives** according to the **project acceptance criteria** within agreed parameters. Project management has **final deliverables** that are **constrained** to a **finite time scale and budget**.

KEY ELEMENTS

- **Scope**
  - Defines project **goals, deliverables, features, and boundaries** (what's in and out).
- **Time**
  - The project **timeline, tasks, milestones, and dependencies**.
- **Cost**
  - Defines project goals, deliverables, features, and boundaries (what's in and out).
- **Quality**
  - The project timeline, tasks, milestones, and dependencies.

B PROJECT LIFE CYCLE

PROJECT LIFE CYCLE

- It is a **five-phase framework** project managers use to **guide** their projects to completion. It **outlines** all **project stages**, from **inception to final implementation**, and describes what the team must **achieve during each phase**.

Summary of the project life cycle phases and their primary objectives.

1. **Initiation**
   - Define the project's **scope, cost, feasibility, goals, timeline**, and **success criteria**.
2. **Planning**
   - Create a project **roadmap** or **detailed action plan** that outlines the tasks needed to complete the project.
3. **Execution**
   - **Carry out the work** to complete the product, service, or result. Ensure your team has all the necessary **resources, tools, and instructions**.
4. **Monitoring and controlling**
   - **Monitor the progress** and **performance** of your **team** and **project** to identify any **deviations** from the plan or **quality issues** so you can make changes accordingly.
5. **Closure**
   - **Close the project**, sign any outstanding contracts, pay vendors, conduct a post-project review, and identify how to improve next time.

C HISTORY BEHIND GANTT CHART

THE HISTORY BEHIND GANTT CHART

- Gantt charts evolved from **Karol Adamiecki**'s late 19th-century **"harmonogram"** for **production scheduling**, but were **popularized** and **named** after **Henry Gantt**, an American engineer who **developed his version** around **1910** to **visualize project tasks** against a timeline for better productivity, notably used in **U.S. war efforts during WWI**.

D GANTT CHART

GANTT CHART

- A Gantt chart is a **project management tool** assisting in the **planning** and **scheduling** of projects of all sizes; they are particularly useful for **visualising projects**. A Gantt chart is defined as a **graphical representation of activity against time**; it helps project professionals **monitor progress**.

GANTT CHART: PURPOSE

- The main purpose of a Gantt chart is to **visually plan, schedule, and track projects** by showing tasks against a **timeline**, allowing managers to see **task durations, dependencies, and progress at a glance**, which aids in resource allocation, risk identification, and clear communication to stakeholders.

D1 KEY COMPONENTS

KEY COMPONENTS OF GANTT CHART

- **Task List**
  - *Vertical axis* showing all project activities
- **Timeline**
  - *Horizontal axis* displaying dates (days, weeks,months)
- **Task Bars**
  - *Horizontal bars* representing task duration
- **Dependencies**
  - *Arrows/lines* connecting related tasks
- **Milestones**
  - *Diamond markers* for key project dates
- **Progress Indicators**
  - *Shaded portions* showing completion status
- **Resource Assignment**
  - *Team members* assigned to tasks

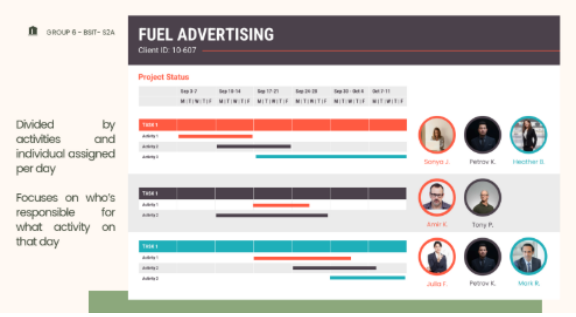
D2 ADVANTAGES & DISADVANTAGES

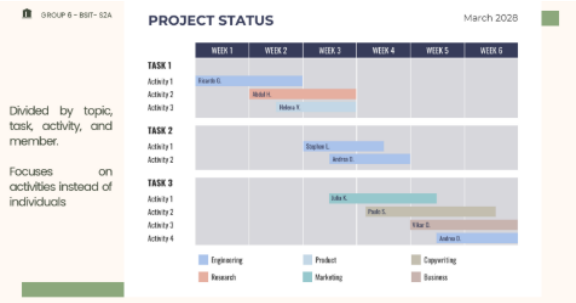
ADVANTAGES

- **Visual Clarity**
  - Easy tounderstandataglance
  - Simplifies complex schedules
- **Better Planning**
  - Shows entire project timeline
  - Identifies potential bottlenecks
- **Track Progress**
  - Monitor which tasks are on schedule
  - Quickly spot delays
- **Improved Communication**
  - Common reference point for teams
  - Clear stakeholder updates

DISADVANTAGES

- **Complexity Issues**
  - Can become cluttered with large projects
  - Difficult to read with 100+ tasks
- **Maintenance Challenges**
  - Requires frequent updates
  - Time-consuming to modify
- **Limited Information**
  - May oversimplify task complexity
  - Doesn't always show resource constraints
  - Critical path not always clear
- **Flexibility**
  - Can be rigid when changes are needed





D3 TOOLS FOR CREATING GANTT CHARTS

TOOLS FOR CREATING GANTT CHARTS

- **MicrosoftExcel**
  - Basic charts, widely accessible
- **MicrosoftProject**
  - Industry standard, full-featured
- **Monday.com**
  - Visualprojectmanagement
- **TeamGantt**
  - Drag-and-drop simplicity
- **Smartsheet**
  - Cloud-based, collaborative
- **GanttProject**
  - Open-source desktop app
- **GoogleSheets**
  - Templates available
- **ClickUp**
  - All-in-one workspace
- **ProjectLibre**
  - Free alternative to MS Project
- **Asana**
  - User-friendly interface

E PROJECT REPORT

PROJECT REPORT

- A **formal document** that **communicates** the **status, progress, and outcomes** of a project to stakeholders
- **Purpose:**
  - Keeps stakeholders informed
  - Documents project activities
  - Records decisions and changes
  - Provides data for future reference

E1 TYPES OF PROJECT REPORTS

TYPES OF PROJECT REPORTS

1. **Status Report**
   - **Customarily floated** periodically to report the **updated state** of the project.
2. **Progress Report**
   - It is a **more specific** report on the events that have been **accomplished** in a **given phase** of the project.
3. **Feasibility Report**
   - Evaluates the **feasibility** and **possibilities** of a suggested **planar strategy**.
4. **Final Report**
   - The project is **closed** at the **end of the performance** and a **comprehensive report** on the project life cycle.
5. **Risk Report**
   - Intelligently **recognizes hazards** that are likely to impede the realization of a given project.
6. **Technical Report**
   - delves into the **technical aspects** of the project, often focusing on **methodologies**, **engineering processes, or scientific data**.

E2 PARTS OF A PROJECT REPORTS

PARTS OF A PROJECT REPORTS

1. **Executive Summary**
   - It is the **concise overview** of the project made to **briefly grasp** the *purpose, key details, findings, methods, and recommendations*.
2. **Introduction**
   - It explains the project's **topic, background, significance**, and what its **purpose** and **limitations**.
3. **Objectives**
   - These are the **specific plans** that defines the **project's goals** and they should be **Specific, Measurable, Achievable, Relevant, and Time-bound (SMART)**.
4. **Methodology**
   - This is the **process** of the project, on how the **information** is being taken.

5. **Findings/Analysis**
    - the process of **analyzing the current status** of a project and **detecting possible issues** as the project advances.
6. **Conclusion**
    - the **summary** of key findings, **answers the hypothesis**, and explains the **implications** of the project while it provides **closure**.
7. **Recommendation**
    - These are the **suggestions** that can be taken ahead of time to **improve the project**, enhance the **functionality** of a project, or to guide any possible work in the future.

F GANTT CHART AND PROJECT REPORT

GANTT CHART AND PROJECT REPORT

- The correlation between Gantt chart and project report is that **Gantt chart is the basis of project report**.
- The Gantt chart is the **main provider of up-to-date information** in the project report. It provides **real-time information** regarding the **progress** of the tasks, milestone achievements, and schedule changes. This current status is then **communicated in a project report** in the form of a **formally structured report**.

F1 PROJECT PLANNING VS PROJECT REPORTING

PROJECT PLANNING

- **Unclear Scope**
    - Leads to uncontrolled changes
- **Unrealistic Schedule**
    - Sets impossible deadlines from the start.
- **Poor Resource Plan**
    - Causes bottlenecks and team overwork.
- **No Risk Planning**
    - Leaves project vulnerable to surprises.
- **Bad Communication Plan**
    - Causes misalignment and confusion.

PROJECT REPORTING

- **Wrong Audience Focus**
    - Report doesn't match stakeholder needs.
- **Only Historical Data**
    - Lacks forecasts and required decisions.
- **Information Overload**
    - Too much data obscures key messages.
- **Inconsistent Format**
    - Changing templates confuse readers.
- **Reports Not Used**
    - Document is filed away, not acted upon.